

$$2x - 1 = \sqrt{4 - 5x}$$

QUESTION 1

1.1 Solve for x:

1.1.1 $x^2 + x - 30 = 0$ (3)

$$(x+6)(x-5) = 0$$

$$\underline{x = -6} \text{ or } \underline{x = 5}$$

$$\begin{array}{c} 30 \\ 15 \\ 10 \\ 6 \\ 5 \\ 3 \\ 2 \\ 1 \end{array}$$

1.1.2 $x(2x - 6) = -3$ (correct to TWO decimal places) (4)

$$2x^2 - 6x + 3 = 0$$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(2)(3)}}{2(2)}$$

$$= \frac{6 \pm \sqrt{12}}{4}$$

$$x = 2,37 \text{ or } 0,63$$

1.1.3 $x^2 - 2x + 1 > 0$ (3)

$$(x-1)(x-1) > 0$$

$$x = 1$$

$$x \in \mathbb{R}, x \neq 1$$

x

$$\begin{aligned} 2x - 1 &= (\sqrt{4 - 5x})^2 \\ 1.1.4 \quad (2x - 1)^2 &= (\sqrt{4 - 5x})^2 \\ 4x^2 - 2x + 1 &= 4 - 5x \\ 4x^2 + x - 3 &= 0 \\ (4x - 3)(x + 1) &= 0 \\ x &= \frac{3}{4} \text{ or } x = -1 \end{aligned}$$

1.2 Solve simultaneously for x and y:

$y - 2x = -1$ and $2y^2 + 4xy = 6x^2$ (6)

$$y^2 + 2xy = 3x^2 \quad \text{--- (2)}$$

$$y = 2x - 1 \quad \text{--- (3)}$$

Sub (3) into (2)

$$(2x - 1)^2 + 2x(2x - 1) = 3x^2$$

$$4x^2 - 4x + 1 + 4x^2 - 2x - 3x^2 = 0$$

$$5x^2 - 6x + 1 = 0$$

$$(5x - 1)(x - 1) = 0$$

$$x = 1 \text{ or } x = \frac{1}{5}$$

$$\text{Sub } x = 1 \text{ or } x = \frac{1}{5}$$

$$y = 2(1) - 1 \text{ or } y = 2\left(\frac{1}{5}\right) - 1$$

$$= 1 \text{ or } y = -\frac{3}{5}$$

1.3 Given the quadratic equation: $2x^2 - px + 1 = 0, x \in \mathbb{R}$.

Determine the possible value(s) of p , such that the equation has two **unequal** real roots.

$$b^2 - 4ac > 0$$

$$(p)^2 - 4(2)(1) > 0$$

$$p^2 - 8 > 0$$

$$p^2 > 8$$

$$p > \sqrt{8} \quad p < -\sqrt{8}$$

Real number
or possible number.

(5)

QUESTION 2

- 2.1 The tenth and the seventeenth terms of an arithmetic sequence are 21 and 49 respectively.

$$\begin{aligned} T_1 &= a \\ T_2 &= a + d \\ T_3 &= a + 2d \end{aligned}$$

- 2.1.1 Determine the common difference of the sequence.

(3)

$$\begin{aligned} a + 9d &= 21 \\ a + 16d &= 49 \end{aligned} \quad \left| \begin{aligned} 7d &= 28 \\ d &= 4 \end{aligned} \right.$$

- 2.1.2 Calculate: $T_1 + T_{18}$ (3)

$$\begin{aligned} T_{10} &= a + 9d = 21 \\ a &= 21 - 4(9) \\ &= -15 \end{aligned}$$

$$\begin{aligned} T_1 + T_{18} &= (a) + (a + 17d) \\ &= -15 + (-15 + 17(4)) \\ &= -15 + 53 \\ &= 38 \end{aligned}$$

- 2.2 Given: $\sum_{n=1}^m (4n - 19) = 1189$
n=1

- 2.2.1 Write down the first three terms of the series. (1)

$$\begin{aligned} T_1 &= 4(1) - 19 = -15 \\ T_2 &= 4(2) - 19 = -11 \\ T_3 &= 4(3) - 19 = -7 \end{aligned}$$

- 2.2.2 Calculate the value of m . (4)

$$\begin{aligned} S_n &= \frac{n}{2} [2a + (n-1)d] \\ S_m &= \frac{m}{2} [2(-15) + 4(m-1)] \\ 1189 &= 2m^2 - 17m \end{aligned}$$

$$\begin{aligned} 0 &= 2m^2 - 17m - 1189 \\ m &= \frac{17 \pm \sqrt{17^2 - 4(2)(-1189)}}{2(2)} \\ m &= 29 \quad m \neq -\frac{41}{2} \end{aligned}$$

- 2.3 -78; -76; -72; -66; ... is a quadratic number pattern.

- 2.3.1 Write down the next two terms of the number pattern. (1)

-78; -76; -72; -66; ... -60; -52

Next 2 $T_n = -60$ and -52

- 2.3.2 Determine the n^{th} term of the number pattern

in the form, $T_n = an^2 + bn + c$.

$$\begin{aligned} 2a &= 2 \\ a &= 1 \end{aligned}$$

$$\begin{aligned} 3a + b &= 2 \\ b &= 2 - 3(1) \\ b &= -1 \end{aligned}$$

$$\begin{aligned} a + b + c &= -78 \\ 1 - 1 + c &= -78 \\ c &= -78 \\ T_n &= n^2 - n - 78 \end{aligned}$$

- 2.3.3 A constant k is added to T_n such that all the terms of the quadratic number pattern become positive.

Determine the value(s) of k . (2)

$k > 78$

[18] QUESTION 3

- 3.1 The first term of a geometric sequence is 81 and the common ratio is r . The sum of the first and third terms of the same geometric sequence is 117. Calculate the value of r .

$$T_n = a(r)^{n-1}$$

$$a = 81 \dots \textcircled{1}$$

$$a + ar^2 = 117 \dots \textcircled{2}$$

$$81 + (81)r^2 = 117$$

$$81r^2 = 36$$

$$r^2 = \frac{36}{81}$$

(4)

$$r = \pm \frac{2}{3}$$

- 3.2 Given the convergent geometric series:
 $3^x + 9^x + 27^x + 81^x + \dots$

- 3.2.1 Write down the common ratio in terms of x . (1)

$$\begin{aligned} r &= \frac{T_2}{T_1} \\ &= \frac{9^x}{3^x} \\ &= \frac{(3 \cdot 3)^{2x}}{3^x} = \frac{3^{2x}}{3^x} = 3^x \end{aligned}$$

- 3.2.2 Calculate the value of x , if $S_\infty = \frac{1}{2}$. (3)

[8]

$$S_\infty = \frac{a}{1-r}$$

$$\frac{1}{2} = \frac{3^x}{1-3^x}$$

$$2 \cdot 3^x = 1 - 3^x$$

$$3 \cdot 3^x = 1$$

$$3^x = \frac{1}{3}$$

$$3^x = 3^{-1}$$

$$x = -1$$

QUESTION 4

Given: $f(x) = \frac{2}{x-5} + 3$

- 4.1 Write down the equations of the asymptotes of f .

$$x - 5 = 0$$

$$x = 5$$

$$y = 3$$

4.2 Write down the range of f .

$$y \neq 3$$

$$y \in \mathbb{R}$$

4.3 Determine the coordinates of the x - and y -intercepts of f . (3)

$$x - 1 = 6$$

$$\frac{2}{x-5} + 3 = 0$$

$$\frac{2}{x-5} = -3$$

$$-3x + 15 = 3$$

$$-3x = -12$$

$$x = 4$$

$$y = -1.66$$

$$y = \frac{2}{-5} + 3$$

$$= \frac{13}{5}$$

4.4 Sketch the graph of f , clearly showing all asymptotes and intercepts with the axes. (4)



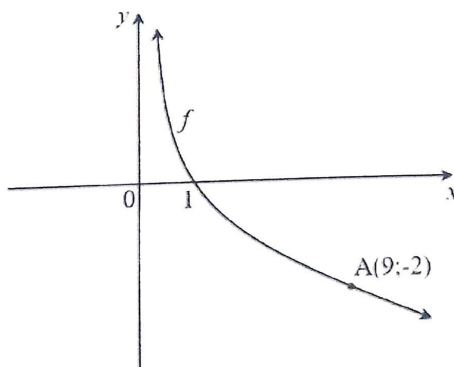
4.5 Describe the transformation that the graph of f has to undergo to form the graph of h , where

$$h(x) = -x^2 - 5$$

2 unit down

QUESTION 5

The diagram below shows the graph of $f(x) = \log_b x$, where b is a constant. f passes through the point $A(9; -2)$.



$$y = a^x$$

$$x = \log_y(a)$$

$$x = 3$$

$$x = \log_y(a)$$

$$a^x = b$$

$$a = \log_x(b)$$

5.1 Show that $b = \frac{1}{3}$.

$$x = 6^y$$

$$9 = 6^{-2}$$

$$6^2 = \frac{1}{9}$$

$$b = \frac{1}{3}$$

5.2 Determine the equation of f^{-1} , the inverse of f , in the form $y = \dots$

$$x = \log_b(y)$$

$$x = \log_{1/3}(y)$$

$$\left(\frac{1}{3}\right)^x = y$$

- 5.3 For which values of x is $f(x) \geq 0$?

$$0 < x \leq 1$$

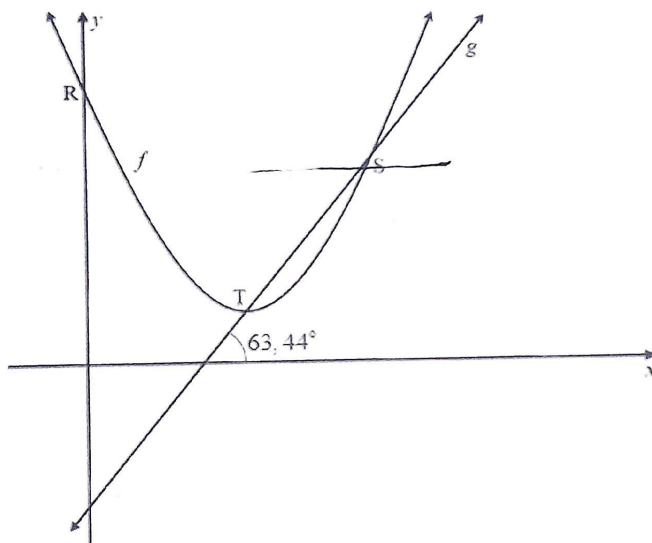
- 5.4 Write down the equation of the asymptote of g , if $g(x) = f^{-1}(x+1)$.

$$g(x) = \left(\frac{1}{3}\right)^{x+1}$$

$$y = 0$$

QUESTION 6

The diagram below shows the graphs of $f(x) = x^2 - 6x + 11$ and $g(x) = ax + b$. The graphs of f and g intersect at S and T , where T the turning point of f . The angle of inclination of g is $63,44^\circ$.



$$6.1 \quad x = \frac{-b}{2a}$$

$$x = \frac{-(-6)}{2(1)} = 3$$

$$y = (3)^2 - 6(3) + 11 = 2$$

$$T(3, 2)$$

- 6.1 Calculate the coordinates of T . (4)

- 6.2 Determine the equation of g in the form $y = mx + c$. (3)

$$m = \tan(63,44^\circ)$$

$$m = 2$$

$$y = 2x + c$$

$$2 = 2(3) + c$$

$$c = -4$$

$$y = 2x - 4$$

- 6.3 Hence, or otherwise determine the coordinates of S . (4)

$$f(x) = g(x)$$

$$x^2 - 6x + 11 = 2x - 4$$

$$x^2 - 8x + 15 = 0$$

$$(x-3)(x-5) = 0$$

$$x = 5 \quad y = 2(5) - 4 = 6$$

$$S(5, 6)$$

- 6.4 Determine the values of:

6.4.1 x , for which $f(x) \leq 6$. (2) $1 \leq x \leq 5$